Breaking Barriers Through Touch: An Affordable Tactile Controller for Vision-Impaired Computer Access

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Abstract

Tangible Accessibility Controller is a low-cost physical control panel designed to help blind, low-vision, and motor-impaired users operate a computer without relying on a keyboard or mouse. The system uses large tactile buttons, a rotary scroll knob, audio feedback, and vibration-based haptic confirmation to enable accessible interaction. An Arduino-based controller sends commands to a laptop, where a Python program executes actions such as reading text aloud, scrolling, and sending preset messages using offline text-to-speech. The design emphasizes simplicity, safety, privacy, and ease of learning, making it suitable for daily use in resource-constrained environments.

Introduction

Many people with visual impairments or limited fine motor control struggle to use traditional input devices such as keyboards and mice. Graphical interfaces are often visually dense, difficult to navigate, and require precise input. Existing assistive technologies are frequently expensive, complex, or dependent on cloud services, raising accessibility and privacy concerns. There is a need for a simple, affordable, and tactile-first solution that enables independent computer use.

Background and Motivation

Accessibility in human–computer interaction is essential for inclusive technology. Blind and low-vision users rely heavily on non-visual cues such as touch and sound. Software-only solutions often lack physical orientation and tactile feedback. This project is motivated by the idea that a physical, touch-based interface combined with clear audio and haptic feedback can significantly reduce interaction barriers while preserving user privacy through offline processing.

Structures of Model

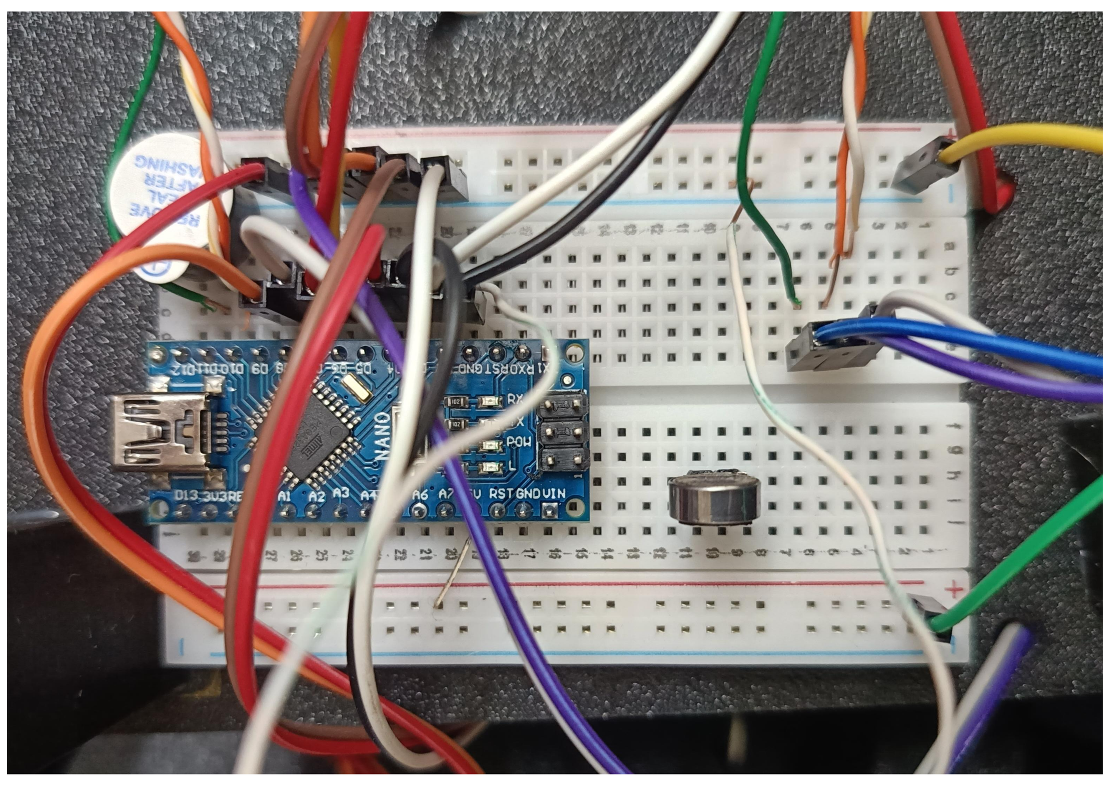
System Overview

The system is a tabletop physical controller consisting of seven large tactile push buttons and one rotary encoder with a built-in push button. The controller connects to a laptop via USB. User inputs are captured by an Arduino Uno, which sends simple serial commands to a Python application running on the computer. Each action produces immediate audio and vibration feedback to confirm successful interaction.

Hardware Design / Architecture

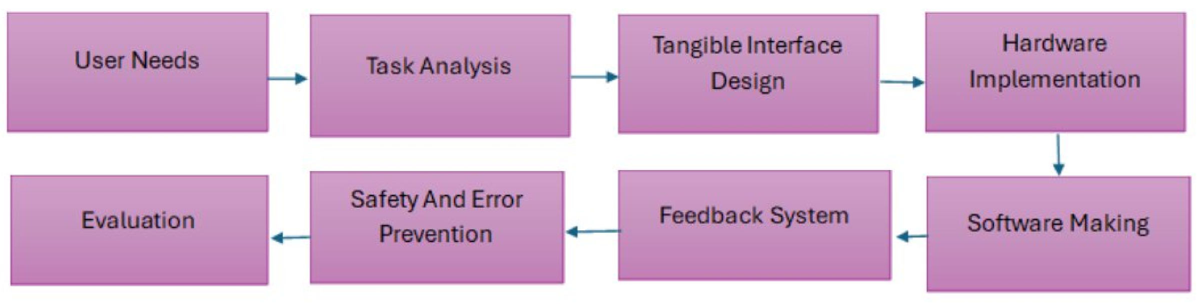
•Arduino Uno R3 as the main controller
•Seven large, uniquely textured push buttons
•One 360° rotary encoder for scrolling and navigation
•Vibration motor for haptic feedback
•USB connection for power and data transfer

The hardware layout is designed to be easily identifiable by touch, with distinct tactile markers and an orientation landmark.



Software Workflow

The Arduino continuously reads button presses and encoder movements and transmits corresponding commands via serial communication. A Python controller program receives these commands and maps them to system actions such as reading clipboard text aloud, scrolling pages, confirming selections, or sending preset messages. Offline text-to-speech ensures fast response and protects user privacy.



Key Features and Innovations

•Tactile-first button design with unique textures and Braille labels
•Combined audio and haptic feedback for every action
•Two-step confirmation for critical actions to prevent accidental input
•Long-press help feature that verbally explains button functions
•Fully offline operation with no cloud dependency
•Low-cost and robust design suitable for daily use

Results / Demonstration

The prototype successfully detected all button presses and encoder rotations with reliable feedback. Scrolling, text reading, and preset message actions worked smoothly in real-time. Users were able to identify buttons by touch and complete basic tasks without visual assistance. The system demonstrated stable performance during live demonstrations.

Table 1: Movement Distances

Movement Path	From Button	To Button	X1,y1	X2,y2	Distance (cm)
Emergency phrase 1	Phrase 1	Ok	(5,12)	(25,12)	20
Emergency phrase 2	Phrase 2	Ok	(11,12)	(25,12)	14
Emergency phrase 3	Phrase 3	Ok	(17,12)	(25,12)	8
Read to back	Read	Back	(8,5)	(25,5)	17
Phrase to Back	Phrase 1	Back	(5,12)	(25,5)	20.88
Read to okay	Read	Ok	(8,5)	(25,12)	18.38
Scroll to Mode	Scroll	mode	(12,19)	(25,19)	13

Table 2: Fitts' Law Index of Difficulty (ID) Calculations

Movement path	Distance D(cm)	Target Width(cm)	D/W Ratio	ID=log2(D/W+1)	Difficulty Rating
Phrase 3->Ok	8	3.3	2.42	1.78	Very Easy
Phrase 2->ok	14	3.3	4.24	2.39	Easy
Phrase 1->ok	20	3.3	6.06	2.82	Moderate
Read->Back	17	3.3	5.15	2.62	Easy
Phrase1->Back	20.88	3.3	6.33	2.87	Moderate
Read->ok	18.38	3.3	5.57	2.72	Easy
Scroll->Mode	13	3.3	3.94	2.30	Easy

Table 3: Predicted Movement Time Using Fitts' Law

Movement path	ID(Index of difficulty)	a(ms)	b(ms)	Tpos(ms)	Time(s)
Phrase 3->Ok	1.78	50	150	317	0.32
Phrase 2->ok	2.39	50	150	409	0.41
Phrase 1->ok	2.82	50	150	473	0.47
Read->Back	2.62	50	150	443	0.44
Phrase1->Back	2.87	50	150	481	0.48
Read->ok	2.72	50	150	458	0.46
Scroll->Mode	2.30	50	150	395	0.40

Discussion

The results indicate that the tangible accessibility controller effectively supports accessible human–computer interaction. Tactile controls, along with audio and haptic feedback, improved user confidence and task performance for reading, scrolling, and selecting options. Core HCI principles, large buttons, and a rotary knob reduced effort and errors, while two-step confirmation minimized accidental actions. Though the prototype currently handles limited tasks and requires initial guidance, it demonstrates the potential of tangible interfaces as practical, low-cost solutions for enhancing accessibility.

Conclusion

[This project demonstrates that an affordable and simple physical controller can greatly improve computer accessibility for visually impaired and motor-impaired users. By combining tactile design, audio feedback, and haptic confirmation, the Tangible Accessibility Controller enables safe, private, and intuitive interaction. The system highlights the effectiveness of physical interfaces in accessible computing.

Future Work

Future improvements include

- adding wireless connectivity
- customizable button mappings
- support for mobile devices, and modular panel layouts tailored to individual user needs.

Reference

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